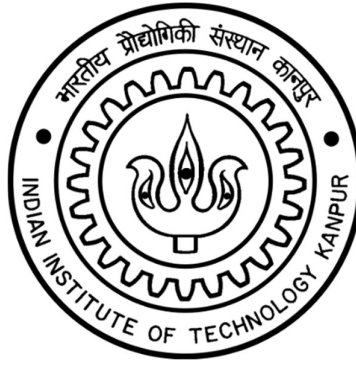


TA 211 PROJECT REPORT

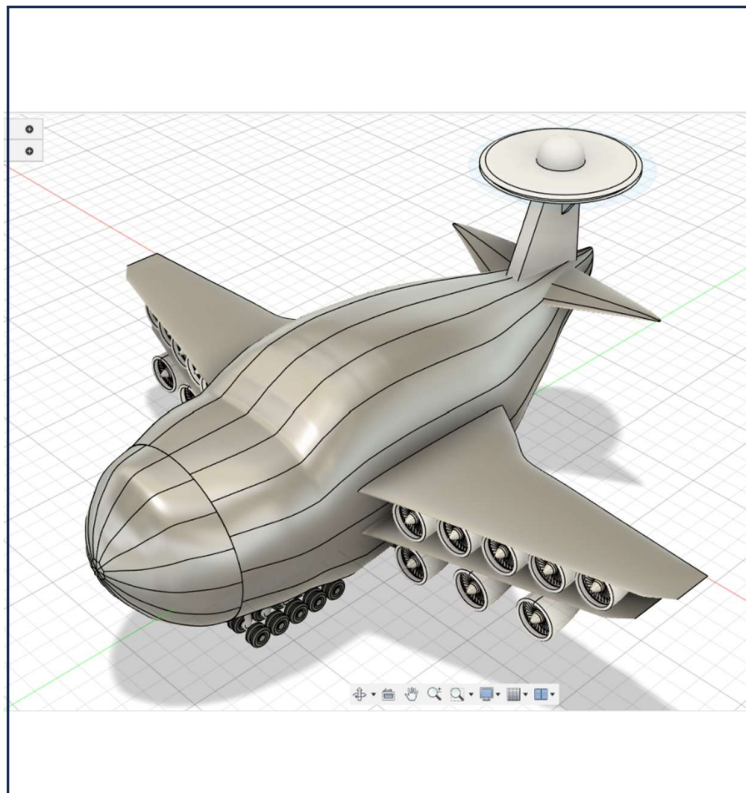


INDIAN INSTITUTE OF TECHNOLOGY KANPUR

COURSE INSTRUCTOR – Prof. Shashank Shekhar

LAB IN-CHARGE – Mr Rakesh Dixit

TUTOR – Prof. Heena Khenchandani



AIR CRUISE

GROUP F14 (FRIDAY)

- 240010 – Aakarsh Omar

- 240235 – Ayan Abbas

- 240285 – Bratadeep Sarkar

- 240483 – Jaiwant N

- 240812 – Priyanshu Yadav

- 240816 – Pushkar Yadav

INDEX

CONTENTS	PAGE NO.
CERTIFICATE OF NON-PLAGIARISM	3
ACKNOWLEDGEMENTS	4
INTRODUCTION	5
MOTIVATION	5
LIST OF PARTS	6
LIST OF MATERIALS	6
MEMBER WISE WORK DISTRIBUTION	7
FULL ASSEMBLY – ISOMETRIC DRAWING	8
FULL ASSEMBLY – ORTHOGRAPHIC DRAWINGS	9-10
SUB ASSEMBLIES – DRAWINGS	11 - 20
PARTS – DRAWINGS	11-20

PLAGIARISM DECLARATION CERTIFICATE

This is to certify that the project report entitled :

“Design and Development of a Scaled Model inspired by the Air Cruise”

Submitted by us in partial fulfilment of the requirements of **Course TA-211**, is a record of the original work carried out by us at **TA-211 Laboratory, DJAB, Indian Institute of Technology Kanpur**.

We hereby declare that:

- The work presented in this report is our own and has not been copied from any source without proper acknowledgement.
- No part of this report has been submitted previously in this or any other institution.
- All sources of information, references, and literature used in the preparation of this report have been duly cited and acknowledged.
- We understand that plagiarism in any form is a serious academic offence, and we take full responsibility for the same.

Signatures of Students:

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_____	240235 – Ayan Abbas
_____	240285 – Bratadeep Sarkar
_____	240483 – Jaiwant N
_____	240812 – Priyanshu Yadav
_____	240816 – Pushkar Yadav

Certification

This is to certify that the above declaration is true to the best of my knowledge and the work has been carried out under my supervision.

Signature: _____

INTRODUCTION

Our project is ‘A scaled model of Air Cruise—an advanced nuclear-powered aircraft-type vessel concept inspired by the “flying hotel” visions in popular science and aerospace engineering.’

What exactly is an Air Cruise nuclear-powered aircraft?

It is an imagined large flying vessel, designed to remain airborne for extended periods as a luxurious hotel and transport hub, powered by a compact nuclear reactor. Unlike conventional aircraft, the Air Cruise is equipped with multi-deck living spaces, viewing domes, leisure facilities, and energy-efficient propulsion, making it a futuristic solution for sustainable, large-scale human travel in the skies.

About Air Cruise –

The Air Cruise concept reimagines aviation by melding elements of airliners, cruise ships, and spacefaring vessels. Proposed as a nuclear-powered aircraft, it would have the capacity to cruise for weeks without refueling, featuring advanced safety systems and multiple redundant engines. Such a vessel might measure hundreds of meters in length, with a spacious central dome, layered wing decks for stability, and large trussed landing gear.

Unlike hydrogen-powered airships, the Air Cruise relies on a reactor-driven electrical system for propulsion, providing consistent long-range capability and unparalleled resilience to weather variations. The luxurious design includes passenger suites, panoramic lounges, service bays, and modular engine pods—emphasizing aerospace engineering, comfort, and safety.

In this project, we aim to construct a scaled physical model of the Air Cruise vessel. The model showcases its multi-wing structure, segmented fuselage, dome canopy, and modular landing gear array. Our approach highlights the engineering and architectural principles underpinning the vessel, serving as both a study in advanced transport design and a visual demonstration tool for next-generation air mobility.

MOTIVATION

Humanity's search for sustainable and innovative long-distance travel has always driven aviation, inspiring new forms of civil and commercial aircraft. The Air Cruise concept represents a leap forward—enabling persistent airborne occupancy, energy independence, and unprecedented comfort for travelers.

Nuclear-powered aircraft, while still conceptual, hold promise for revolutionizing air transport, overcoming the fuel limitations of conventional jets and providing a template for long-duration missions. Exploring this design helps highlight the integration of nuclear energy, safety systems, modular construction, and luxury accommodations—a compelling engineering blend.

This project is motivated by a desire to understand the engineering, design philosophies, and systemic modularity involved in building such an advanced airborne vessel. The construction of a scaled model allows for hands-on exploration of modular structures, distributed propulsion, and layered support systems—making the complexity tangible and accessible. It gives our team insight into fabrication, assembly, and structural problem-solving typical of interdisciplinary aerospace projects.

Another primary motivation is educational. Making a physical Air Cruise model provides an effective means of demonstrating cutting-edge transport concepts to students, faculty, and visitors. The model turns abstract blueprints into practical learning experiences about propulsion, architecture, and safety in next-generation aircraft. For our team, the project also develops practical fabrication and teamwork skills, vital for future roles in aerospace and engineering.

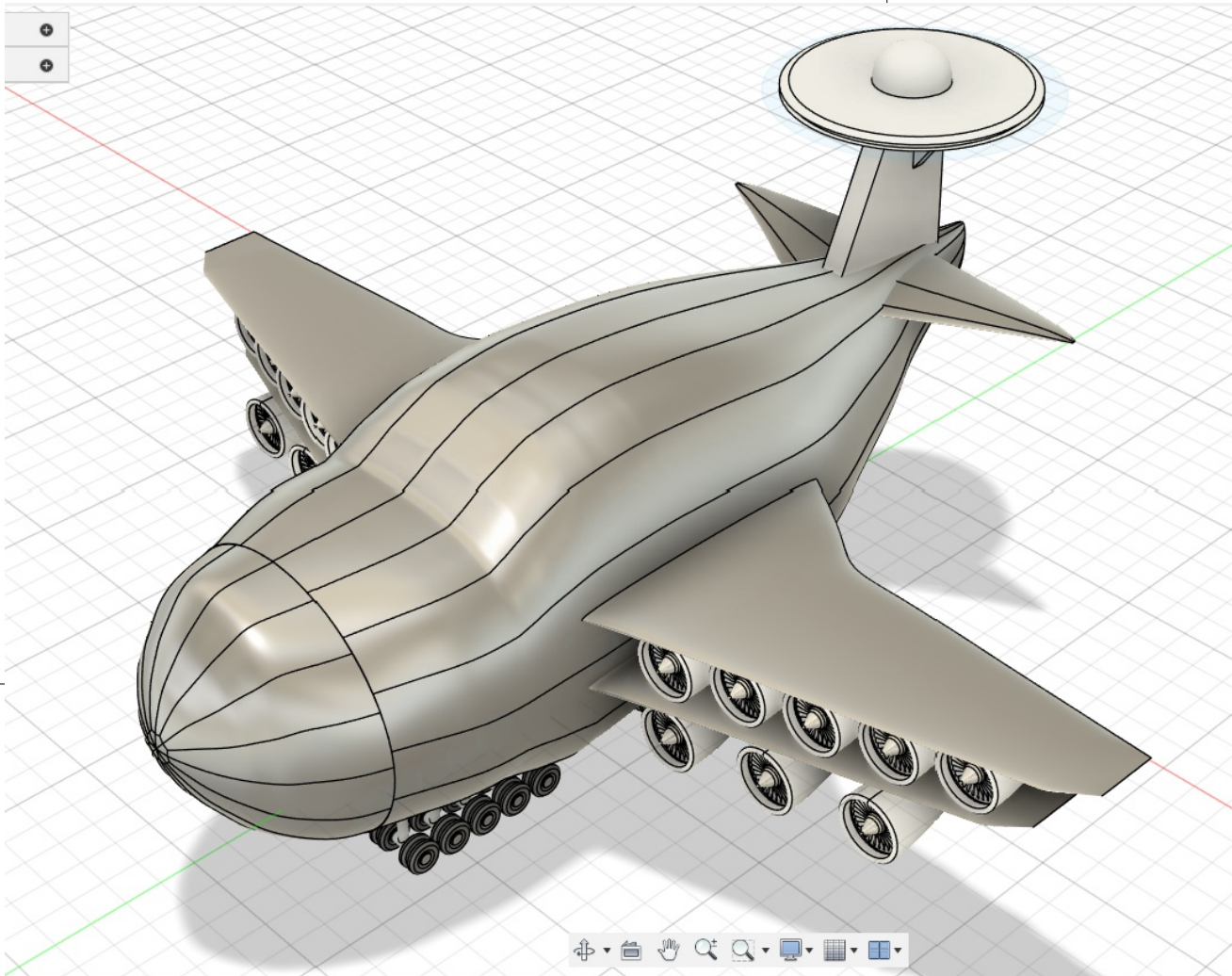
Ultimately, this project reflects our enthusiasm for advanced aerospace technology and sustainable travel. Air Cruise embodies forward-thinking transport, combining engineering marvels with hospitality and safety. By recreating and detailing the vessel at scale, we aspire not only to honor its conceptual brilliance but also to ignite further curiosity and innovation in the field of airborne design and nuclear-powered transport. This motivation underpins our focus on crafting an accurate, detailed model that brings the Air Cruise vision vividly to life.

LIST OF PARTS AND MATERIALS REQUIRED

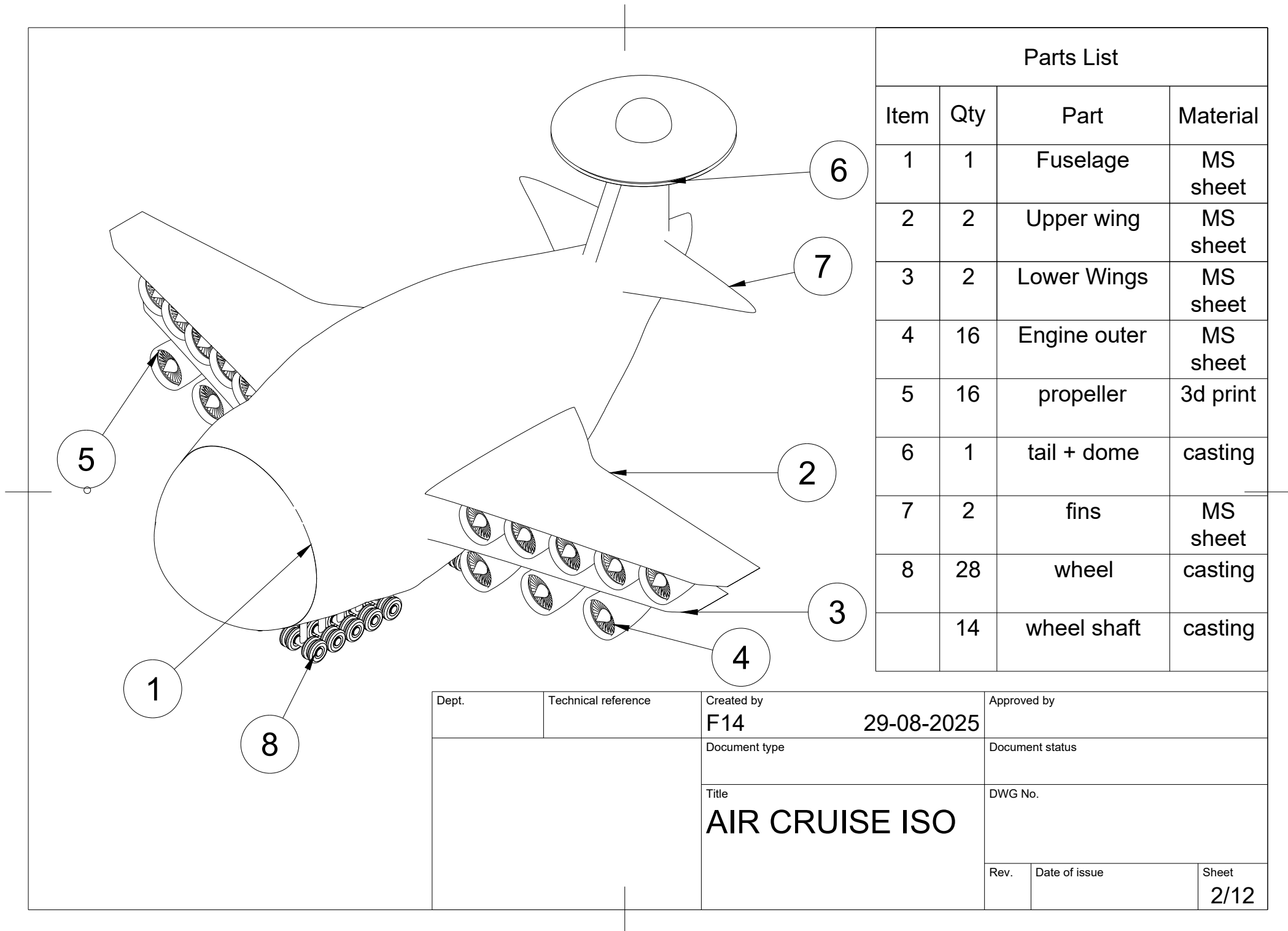
PART NAME	QUANTITY REQUIRED	MATERIALS REQUIRED (PER SINGLE UNIT) (in mm)	Rate (rs)	MANUFACTURING PROCESSES	Total Cost (in Rs)
fuselage	1	1 x (Mild steel sheet - 370 x 370 x 2) (10545.296+70916.642=81461.938)	3090/sheet	Sheet Cutting Bending	3090
upper wings	2	2 x (Mild steel sheet - 100 x 150 x 2)	3090/sheet	Sheet Cutting Bending	3090
lower wings	2	2 x (Mild steel sheet - 100 x 130 x 2)	3090/sheet	Sheet Cutting Bending	3090
fins	2	2 x (Mild steel sheet - 60 x 75 x 1)	1278/sheet	Sheet Cutting Bending	1278
dome	1	1 x (Cast iron – 4019.39 mm ³) 1 x (Thermocol - 100 x 100 x 20)	75/kg	casting	13.60
engine	16	1 x (Mild steel sheet- 40 x 70 x 1)	1278/sheet	Sheet cutting	1278
Propeller	16	1x(Mild steel sheet – 20x20x2) 1x(mild steel sheet – 12x25x1)	3090/sheet 1278/sheet	Sheet cutting	4368
tail	1	2 x (Mild steel sheet - 60 x 75 x 2) 2x(Mild steel sheet – 50x40x2)	3090/sheet 3090/sheet	Sheet Cutting Bending	3090
wheel	28	(10.5 outer dia, 4 inner dia) (Thermacol – 15x15x10) 2*(Cast iron-220mm ³)	75/kg	casting	872
shaft	14	1 x (Mild steel round rod –length 19.5, 4 mm dia) 1x(Mild steel round rod –length 13.13, 4mm dia)	70/kg 70/kg	Tube cutting	1.88

MEMBER WISE WORK DISTRIBUTION

MEMBER NAME	P1	P2	P3	P4	P5	P6
BRATADEEP						
AYAN						
PRIYANSHU						
AAKARSH						
JAIWANT						
PUSHKAR						

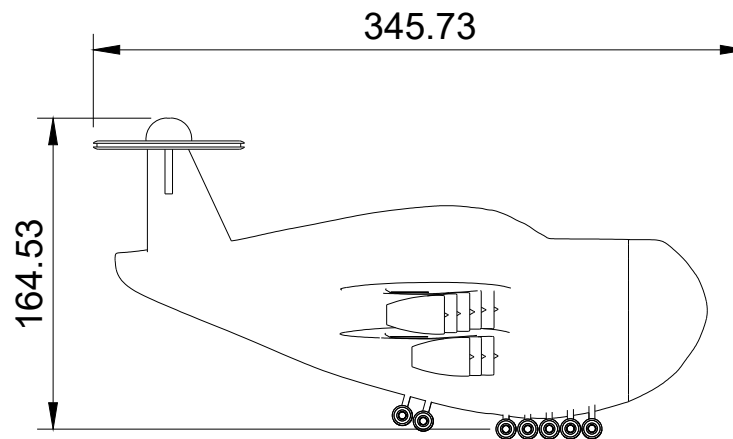
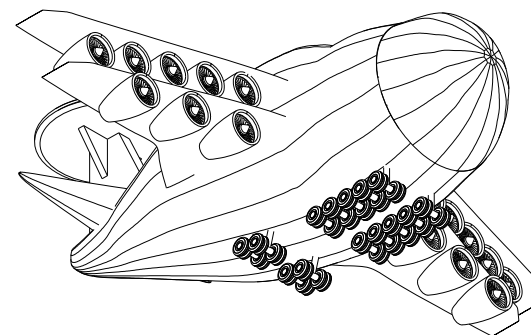
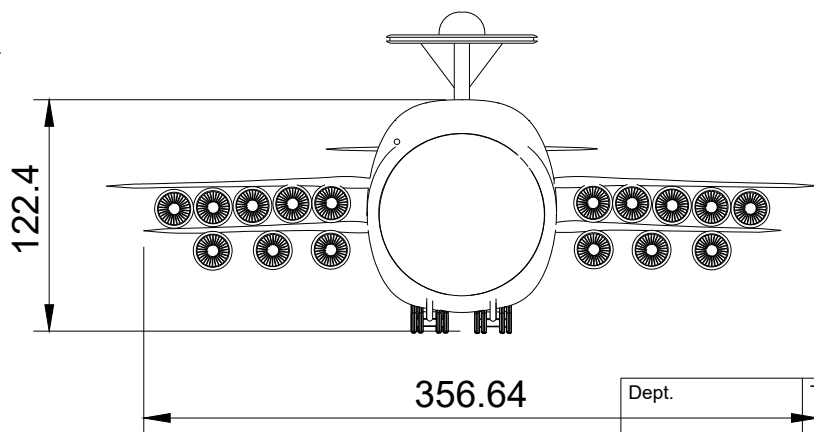
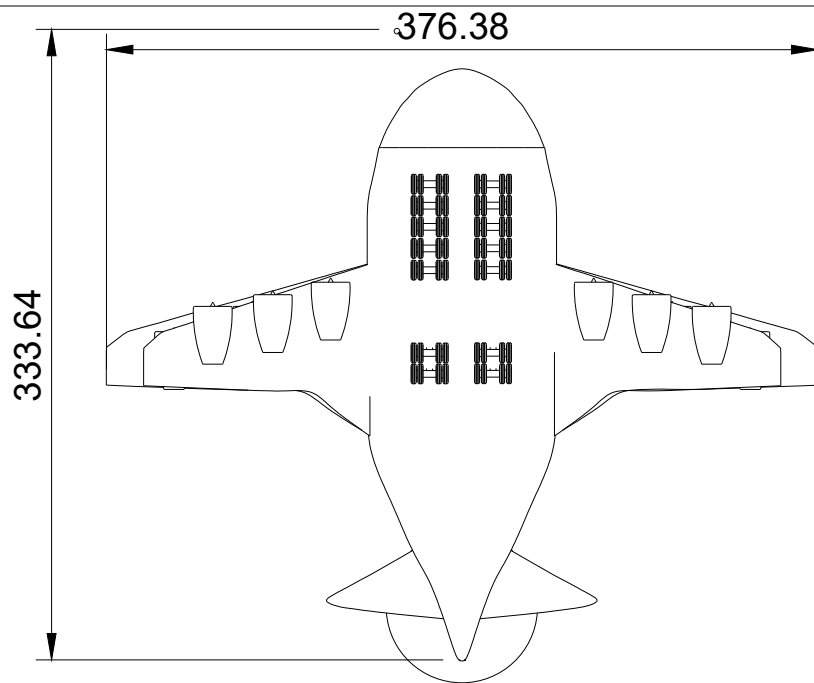


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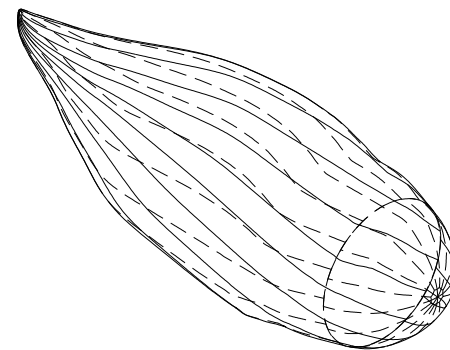
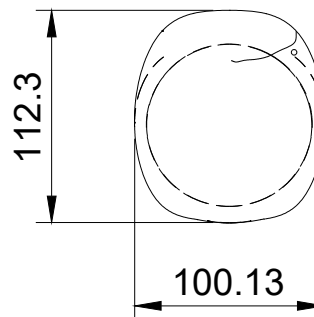
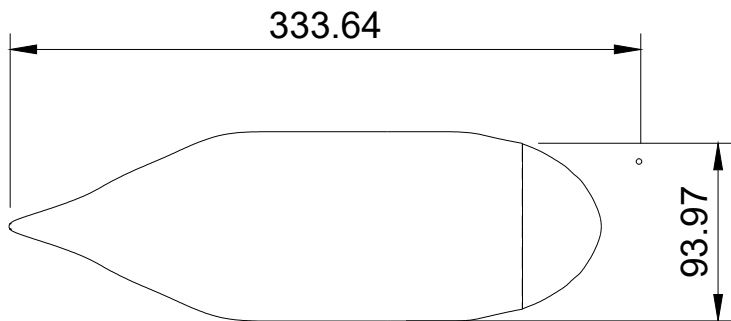


Parts List			
Item	Qty	Part	Material
1	1	Fuselage	MS sheet
2	2	Upper wing	MS sheet
3	2	Lower Wings	MS sheet
4	16	Engine outer	MS sheet
5	16	propeller	3d print
6	1	tail + dome	casting
7	2	fins	MS sheet
8	28	wheel	casting
	14	wheel shaft	casting

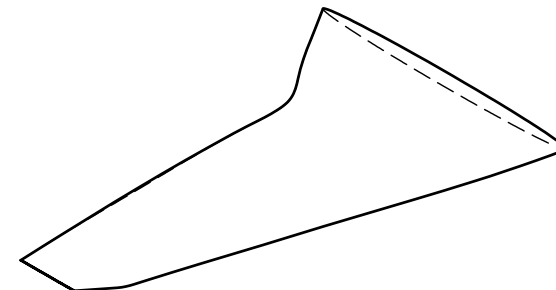
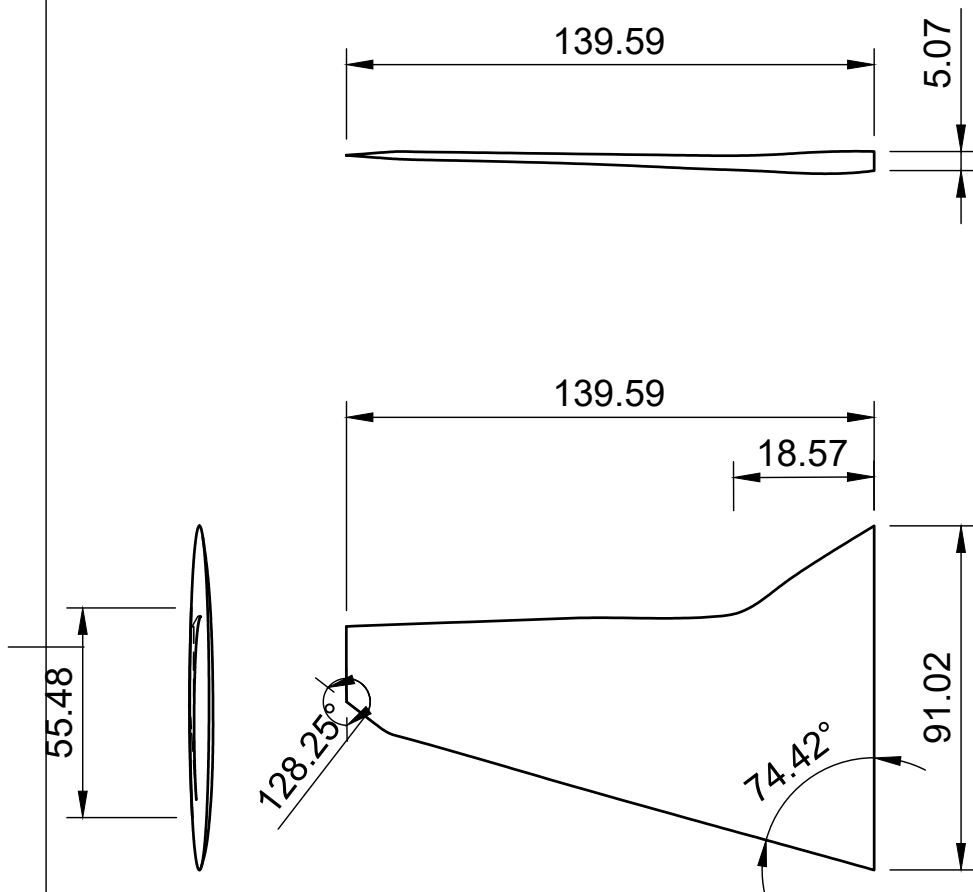
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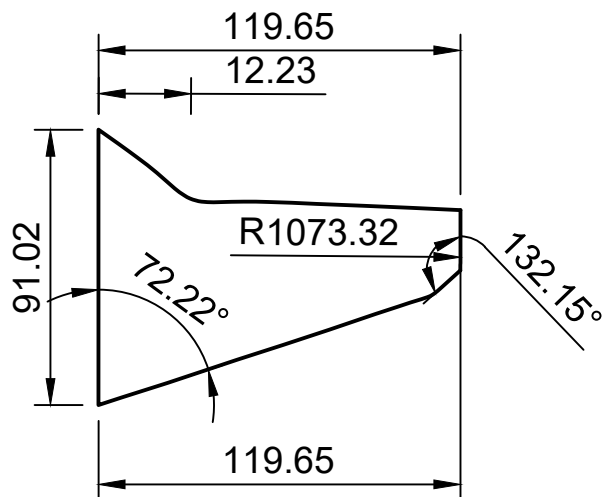
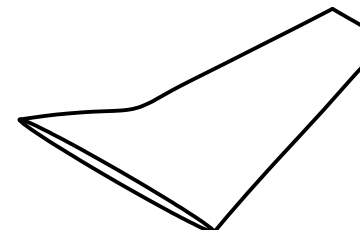
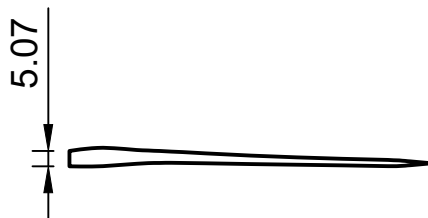
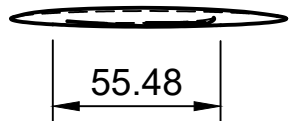
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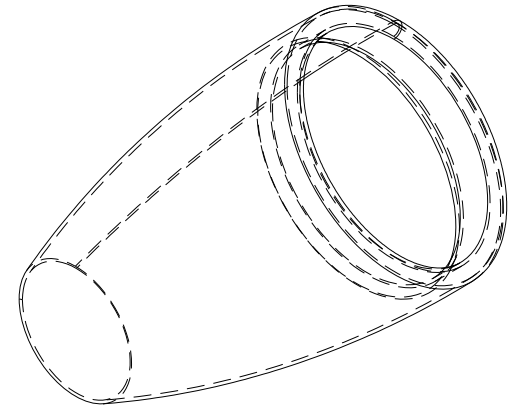
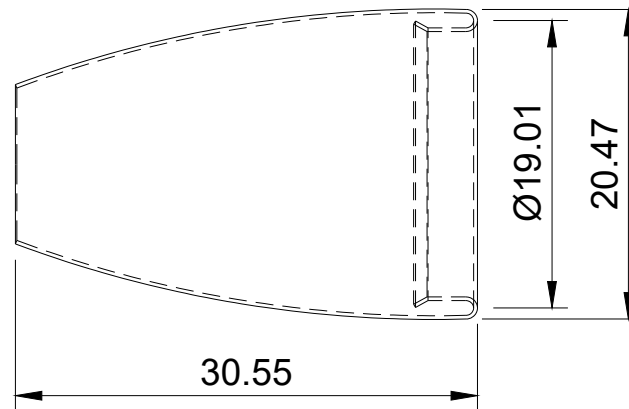
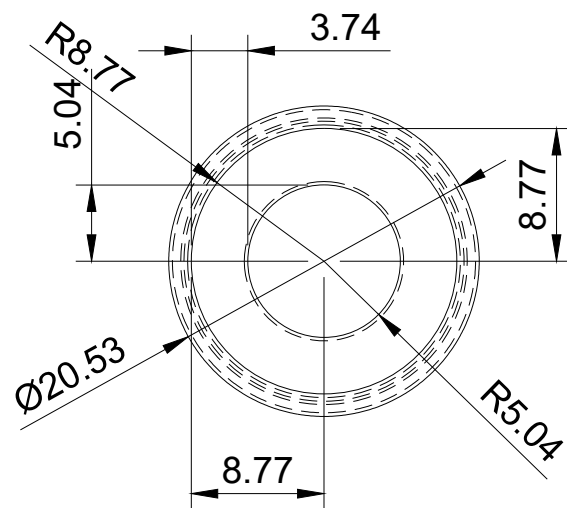
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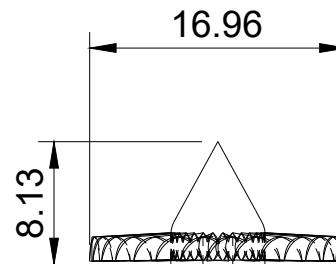
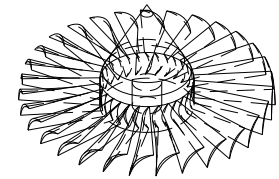
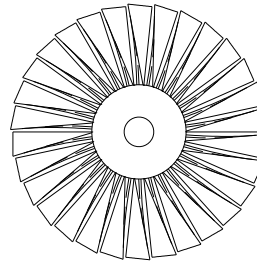
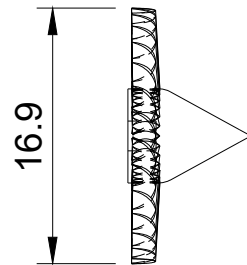
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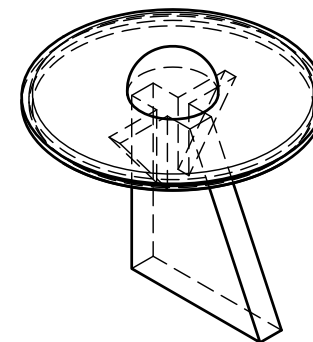
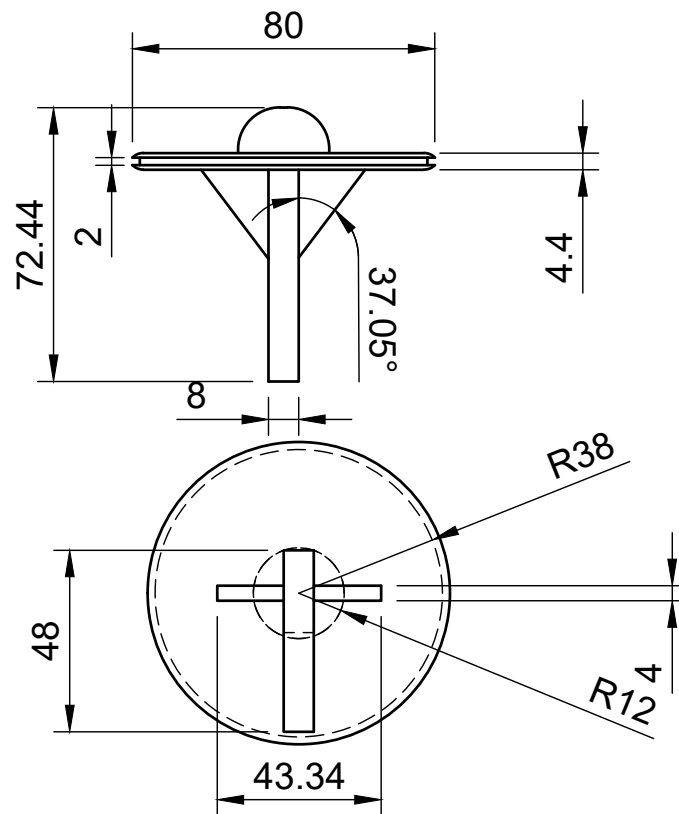
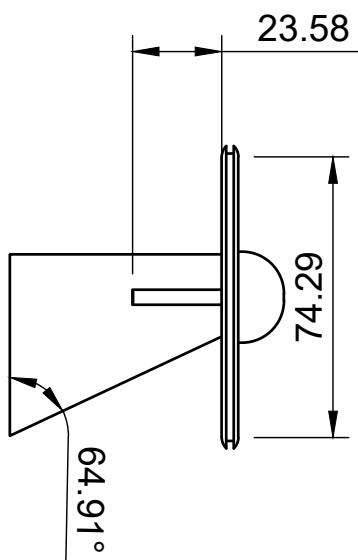
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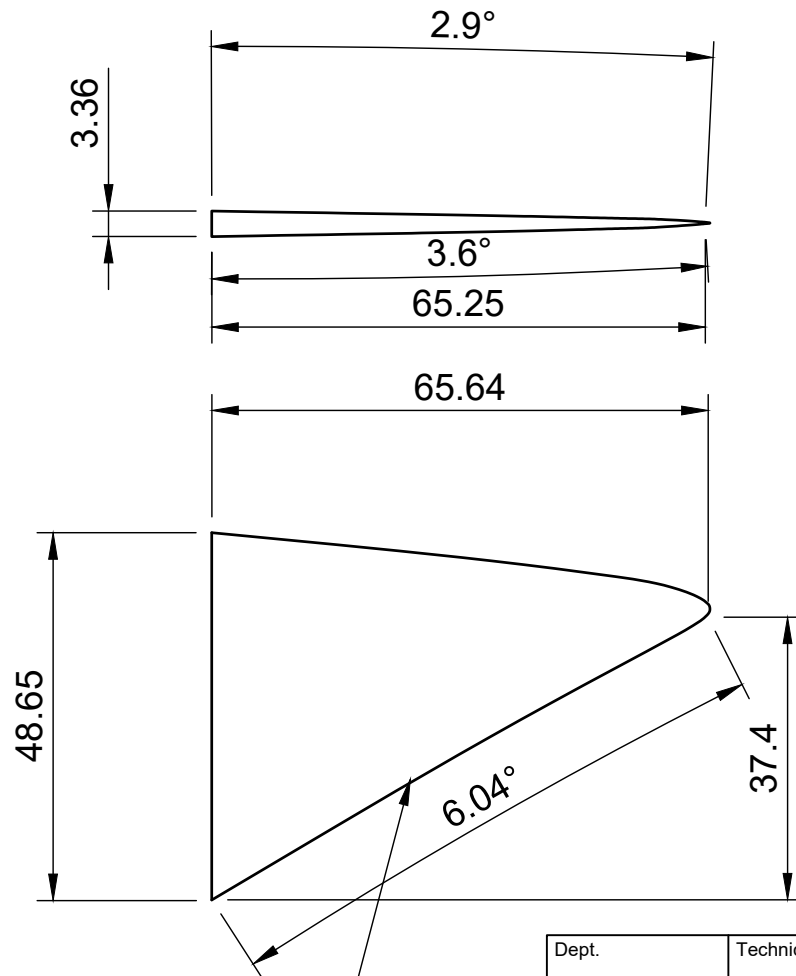
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		Rev.	Date of issue	Sheet 7/12



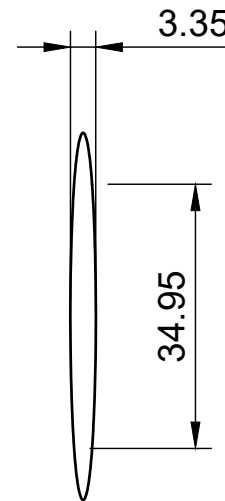
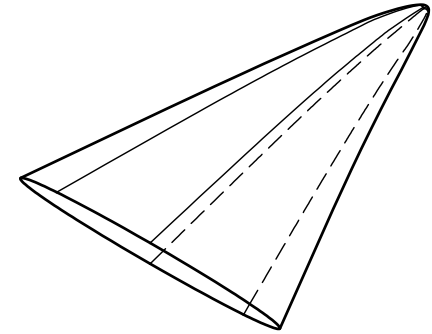
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			Rev.	Date of issue		Sheet 8/12	



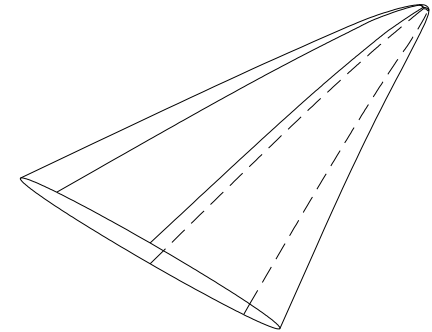
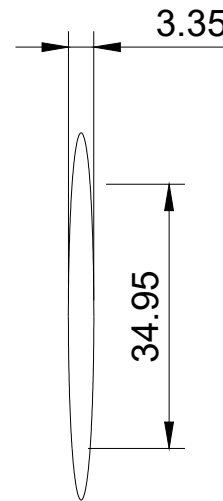
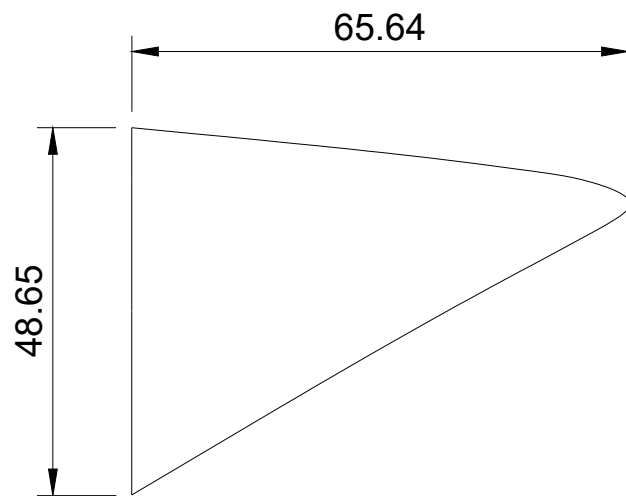
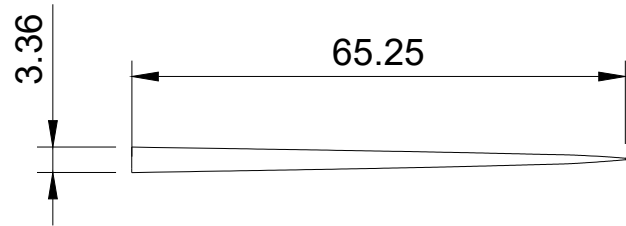
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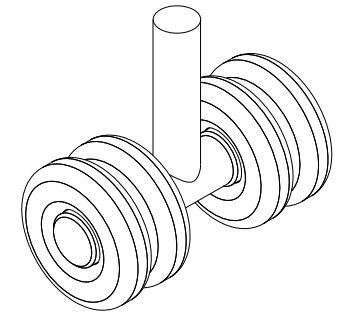
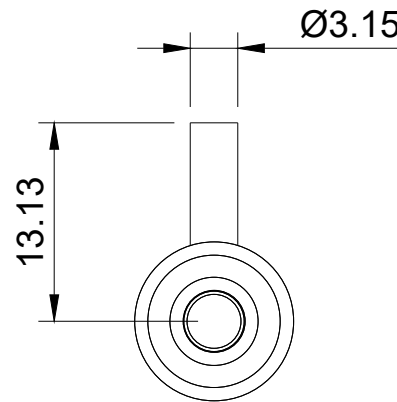
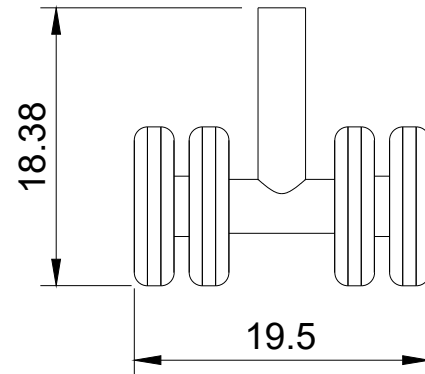
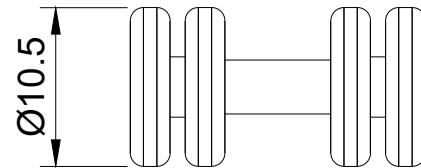
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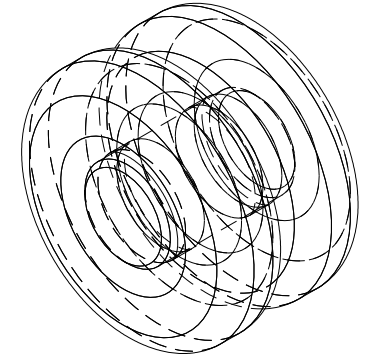
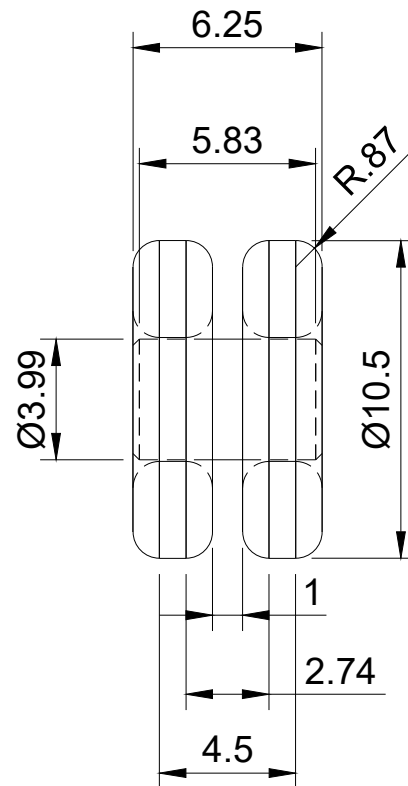
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			Rev.	Date of issue	Sheet 10/12



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		Document type			Document status			
		Title wheel + shaft			DWG No.			
					Rev.	Date of issue	Sheet 11/12	



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		Title wheel				DWG No.	
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